Project Advanced SQL.

Database about fire department management system

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The database represents a fire department management system. It contains information about fire stations, firefighters, vehicles, interventions, and participation in interventions.

Tables and their purpose:

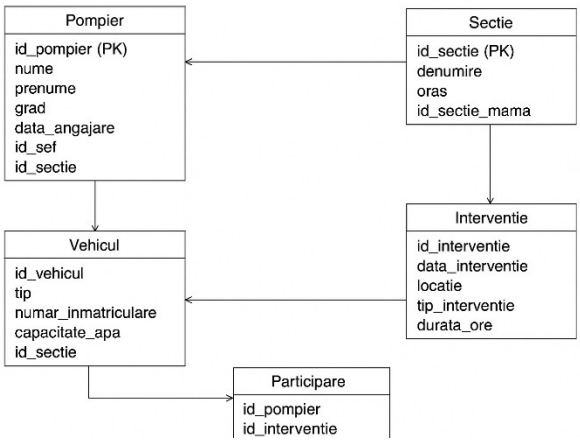
Sectie – Stores information about each fire station, including its name, city, and parent station (for hierarchical structure).

Pompier – Stores personal details of firefighters, their rank, hire date, their manager (if any), and the station they belong to.

Vehicul – Stores information about vehicles available in each station, including type, license plate, and water capacity.

Interventie – Stores details of emergency interventions, such as date, location, type, and duration.

Participare – Connects firefighters to interventions and defines their role in each intervention.



```
CREATE TABLE Sectie (
    id_sectie INT PRIMARY KEY,
    denumire VARCHAR2(50),
    oras VARCHAR2(50),
    id_sectie_mama INT,
    FOREIGN KEY (id_sectie_mama) REFERENCES Sectie(id_sectie)
);

CREATE TABLE Pompier (
    id_pompier INT PRIMARY KEY,
    nume VARCHAR2(50),
    prenume VARCHAR2(50),
    grad VARCHAR2(30),
    data_angajare DATE,
    id_sef INT,
    id_sectie INT,
    FOREIGN KEY (id_sef) REFERENCES Pompier(id_pompier),
    FOREIGN KEY (id_sectie) REFERENCES Sectie(id_sectie)
);

CREATE TABLE Vehicul (
    id_vehicul INT PRIMARY KEY,
    tip VARCHAR2(50),
    numar_inmatriculare VARCHAR2(20),
    capacitate_apa INT,
    id_sectie INT,
    FOREIGN KEY (id_sectie) REFERENCES Sectie(id_sectie)
);
```

```
CREATE TABLE Interventie (
   id_interventie INT PRIMARY KEY,
   data_interventie DATE,
   locatie VARCHAR2(100),
   tip_interventie VARCHAR2(50),
   durata_ore NUMBER(5,2)
);

CREATE TABLE Participare (
   id_pompier INT,
   id_interventie INT,
   rol VARCHAR2(30),
   PRIMARY KEY (id_pompier, id_interventie),
   FOREIGN KEY (id_pompier) REFERENCES Pompier(id_pompier),
   FOREIGN KEY (id_interventie) REFERENCES Interventie(id_interventie)
);
```

The creation of the populated tables

```
INSERT INTO Sectie VALUES (1, 'Sectia Centrala', 'Bucuresti', NULL);
INSERT INTO Sectie VALUES (2, 'Sectia Nord', 'Bucuresti', 1);
INSERT INTO Sectie VALUES (3, 'Sectia Sud', 'Constanta', 1);
INSERT INTO Sectie VALUES (4, 'Punct de lucru Mamaia', 'Constanta', 3);
INSERT INTO Sectie VALUES (5, ’Brasov’, 'Brasov', 1);

INSERT INTO Pompier VALUES (1, 'Popescu', 'Ion', 'Capitan', DATE '2015-03-12', NULL, 1);
INSERT INTO Pompier VALUES (2, 'Ionescu', 'Mihai', 'Locotenent', DATE '2018-05-10', 1, 2);
INSERT INTO Pompier VALUES (3, 'Marin', 'Alexandru', 'Sergent', DATE '2020-07-01', 2, 4);
INSERT INTO Pompier VALUES (4, 'Georgescu', 'Andrei', 'Locotenent', DATE '2019-09-15', 1, 3);
```

```
INSERT INTO Pompier VALUES (5, 'Manea', 'Alina', 'Pompier', DATE '2020-08-15', 4, 3);
INSERT INTO Pompier VALUES (6, 'Manea', 'Claudia', 'Capitan', DATE '2017-05-25', 1, 5);
INSERT INTO Pompier VALUES (7, 'Manole', 'Paul', 'Pompier', DATE '2020-01-15', 1, 1);
INSERT INTO Pompier VALUES (8, 'Pruna', 'Dimitria', 'Pompier', DATE '2022-07-19', 6, 5);
INSERT INTO Pompier VALUES (9, 'Fodor', 'Dimitria', 'Pompier', DATE '2022-07-19', 6, 5);
INSERT INTO Pompier VALUES (10, 'Dumitrescu', 'Andreea', 'Pompier', DATE '2021-02-12', 2, 2);
INSERT INTO Pompier VALUES (11, 'Neagu', 'Mihnea', 'Locotenent', DATE '2016-06-20', 1, 1);
INSERT INTO Pompier VALUES (12, 'Stan', 'Elena', 'Sergent', DATE '2019-11-01', 2, 2);
INSERT INTO Pompier VALUES (13, 'Radu', 'Alex', 'Pompier', DATE '2022-01-05', 4, 3);
INSERT INTO Pompier VALUES (14, 'Iacob', 'Simona', 'Pompier', DATE '2020-05-23', 4, 4);
INSERT INTO Pompier VALUES (15, 'Mihai', 'Bogdan', 'Capitan', DATE '2015-10-12', 1, 1);
INSERT INTO Pompier VALUES (16, 'Popa', 'Cristina', 'Sergent', DATE '2018-03-17', 15, 1);
INSERT INTO Pompier VALUES (17, 'Petrescu', 'Alin', 'Pompier', DATE '2021-07-11', 15, 1);
INSERT INTO Pompier VALUES (18, 'Marinescu', 'Vlad', 'Locotenent', DATE '2017-09-19', 1, 2);
INSERT INTO Pompier VALUES (19, 'Alexandru', 'Laura', 'Pompier', DATE '2022-03-08', 18, 2);
INSERT INTO Pompier VALUES (20, 'Vasilescu', 'Andrei', 'Sergent', DATE '2019-02-27', 11, 1);
INSERT INTO Pompier VALUES (21, 'Enache', 'Ana', 'Pompier', DATE '2020-12-05', 11, 1);
INSERT INTO Pompier VALUES (22, 'Cojocaru', 'Robert', 'Pompier', DATE '2021-08-14', 18, 2);
INSERT INTO Pompier VALUES (23, 'Dobre', 'Ioana', 'Pompier', DATE '2020-11-23', 18, 2);
INSERT INTO Pompier VALUES (24, 'Barbu', 'Radu', 'Capitan', DATE '2016-05-16', 1, 3);
INSERT INTO Pompier VALUES (25, 'Constantinescu', 'Elena', 'Sergent', DATE '2019-04-22', 24, 3);
INSERT INTO Pompier VALUES (26, 'Florea', 'Mihai', 'Pompier', DATE '2021-06-09', 24, 3);
INSERT INTO Pompier VALUES (27, 'Tudor', 'Cristina', 'Pompier', DATE '2022-01-12', 24, 4);
INSERT INTO Pompier VALUES (28, 'Vlad', 'Alina', 'Sergent', DATE '2020-09-14', 4, 4);
INSERT INTO Pompier VALUES (29, 'Lazar', 'Paul', 'Pompier', DATE '2022-05-18', 4, 3);
INSERT INTO Pompier VALUES (30, 'Ionescu', 'Adrian', 'Pompier', DATE '2021-03-21', 4, 3);
INSERT INTO Pompier VALUES (31, 'Pop', 'Andreea', 'Pompier', DATE '2022-08-09', 6, 5);
INSERT INTO Pompier VALUES (32, 'Vasiliu', 'Dumitru', 'Pompier', DATE '2021-10-10', 6, 5);
INSERT INTO Pompier VALUES (33, 'Georgescu', 'Ana', 'Sergent', DATE '2019-12-15', 6, 5);
```

INSERT INTO Pompier VALUES (34, 'Dima', 'Roxana', 'Pompier', DATE '2020-07-19', 6, 5);

INSERT INTO Pompier VALUES (35, 'Marin', 'Victor', 'Pompier', DATE '2022-02-22', 6, 5);

INSERT INTO Vehicul VALUES (1, 'Autospeciala pompieri', 'B123ABC', 3000, 1);

INSERT INTO Vehicul VALUES (2, 'Ambulanta', 'B124ABC', 1000, 2);

INSERT INTO Vehicul VALUES (3, 'Motopompa', 'CT567XY', 500, 4);

INSERT INTO Vehicul VALUES (4, 'Ambulanta', 'B125ABC', 1000, 3);

INSERT INTO Vehicul VALUES (5, 'Autospeciala cu scară', 'CT568XY', 2000, 4);

INSERT INTO Vehicul VALUES (6, 'Motopompa portabilă', 'BV101BV', 500, 5);

INSERT INTO Interventie VALUES (1, DATE '2025-03-01', 'Bucuresti, Str. Victoriei', 'Incendiu clădire', 3.5);

INSERT INTO Interventie VALUES (2, DATE '2025-03-02', 'Constanta, Mamaia', 'Inundatie', 5.0);

INSERT INTO Interventie VALUES (3, DATE '2025-03-03', 'Brasov, Str. Mare', 'Incendiu pădure', 7.0);
INSERT INTO Interventie VALUES (4, DATE '2025-03-04', 'Bucuresti, Str. Libertatii', 'Incendiu apartament', 2.0);

INSERT INTO Interventie VALUES (5, DATE '2025-03-05', 'Bucuresti, Str. Unirii', 'Accident auto', 1.5);

INSERT INTO Interventie VALUES (6, DATE '2025-03-06', 'Constanta, Str. Independentei', 'Inundatie', 4.0);

INSERT INTO Interventie VALUES (7, DATE '2025-03-07', 'Mamaia, Str. Plajei', 'Incendiu magazin', 3.0);

INSERT INTO Interventie VALUES (8, DATE '2025-03-08', 'Brasov, Str. Republicii', 'Incendiu padure', 6.0);

INSERT INTO Interventie VALUES (9, DATE '2025-03-09', 'Bucuresti, Str. Aviatorilor', 'Accident chimic', 5.0);

INSERT INTO Interventie VALUES (10, DATE '2025-03-10', 'Constanta, Str. Tomis', 'Incendiu garaj', 2.5);

INSERT INTO Interventie VALUES (11, DATE '2025-03-11', 'Bucuresti, Str. Plevnei', 'Inundatie', 3.5);

INSERT INTO Interventie VALUES (12, DATE '2025-03-12', 'Brasov, Str. Barsei', 'Accident rutier', 1.0);

INSERT INTO Interventie VALUES (13, DATE '2025-03-13', 'Bucuresti, Str. Calea Victoriei', 'Incendiu bloc', 4.0);

INSERT INTO Interventie VALUES (14, DATE '2025-03-14', 'Mamaia, Str. Lacului', 'Accident auto', 2.0);

INSERT INTO Interventie VALUES (15, DATE '2025-03-15', 'Constanta, Str. Marinarilor', 'Inundatie', 5.5);

INSERT INTO Participare VALUES (1, 1, 'Coordonator');

INSERT INTO Participare VALUES (2, 1, 'Echipă');

INSERT INTO Participare VALUES (3, 2, 'Echipă');

INSERT INTO Participare VALUES (5, 3, 'Echipă');

INSERT INTO Participare VALUES (10, 4, 'Echipă');

INSERT INTO Participare VALUES (11, 5, 'Coordonator');

INSERT INTO Participare VALUES (12, 6, 'Echipă');

INSERT INTO Participare VALUES (13, 7, 'Echipă');

INSERT INTO Participare VALUES (14, 8, 'Echipă');

2 hierarchical queries

1. Hierarchy of stations

SELECT LPAD(' ', LEVEL*2) || denumire AS sectie, oras, id_sectie_mama

FROM Sectie

START WITH id_sectie_mama IS NULL

CONNECT BY PRIOR id_sectie = id_sectie_mama;

```
//ierarhia sectilor
SELECT LPAD(' ', LEVEL*2) || denumire AS sectie, oras, id_sectie_mama
FROM Sectie
START WITH id_sectie_mama IS NULL
CONNECT BY PRIOR id_sectie = id_sectie_mama;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 5 in 0.007 seconds

	SECTIE	ORAS	ID_SECTIE_MAMA
1	Sectia Centrala	Bucuresti	(null)
2	Sectia Nord	Bucuresti	1
3	Sectia Sud	Constanta	1
4	Punct de lucru Mamaia	Constanta	3
5	Brasov	Brasov	1

2. Hierarchy of firefighters

SELECT LPAD(' ', LEVEL*2) || nume || ' ' || prenume AS pompier, grad, id_sef, level

FROM Pompier

START WITH id_sef IS NULL

CONNECT BY PRIOR id_pompier = id_sef

order by level;

```
//ierarhia pompierilor
SELECT LPAD(' ', LEVEL*2) || nume || ' ' || prenume AS pompier, grad, id_sef, level
FROM Pompier
START WITH id_sef IS NULL
CONNECT BY PRIOR id_pompier = id_sef
order by level;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 35 in 0.014 seconds

	POMPIER	GRAD	ID_SEF	LEVEL
1	Popescu Ion	Capitan	(null)	1
2	Ionescu Mihai	Locotenent	1	2
3	Barbu Radu	Capitan	1	2
4	Manea Claudia	Capitan	1	2
5	Manole Paul	Pompier	1	2
6	Marinescu Vlad	Locotenent	1	2
7	Georgescu Andrei	Locotenent	1	2
8	Mihai Bogdan	Capitan	1	2
9	Neagu Mihnea	Locotenent	1	2
10	Pruna Dimitria	Pompier	6	3
11	Fodor Dimitria	Pompier	6	3
12	Pop Andreea	Pompier	6	3

6 queries with various types of joins, subqueries, aggregations

1.Pompieri și secția lor:

SELECT p.nume, p.prenume, p.grad, s.denumire AS sectie

FROM Pompier p

JOIN Sectie s ON p.id_sectie = s.id_sectie;

```
//Pompieri și secția lor:
SELECT p.nume, p.prenume, p.grad, s.denumire AS sectie
FROM Pompier p
JOIN Sectie s ON p.id_sectie = s.id_sectie;
```

Script Output | Query Result

All Rows Fetched: 35 in 0.011 seconds

	NUME	PRENUME	GRAD	SECTIE
1	Popescu	Ion	Capitan	Sectia Centrala
2	Ionescu	Mihai	Locotenent	Sectia Nord
3	Marin	Alexandru	Sergent	Punct de lucru Mamaia
4	Georgescu	Andrei	Locotenent	Sectia Sud
5	Manea	Alina	Pompier	Sectia Sud
6	Manea	Claudia	Capitan	Brasov
7	Manole	Paul	Pompier	Sectia Centrala
8	Pruna	Dimitria	Pompier	Brasov
9	Fodor	Dimitria	Pompier	Brasov
10	Dumitrescu	Andreea	Pompier	Sectia Nord
11	Neagu	Mihnea	Locotenent	Sectia Centrala
12	Stan	Elena	Sergent	Sectia Nord

2. Număr de pompieri per secție:

SELECT s.denumire, COUNT(p.id_pompier) AS numar_pompieri

FROM Sectie s

LEFT JOIN Pompier p ON s.id_sectie = p.id_sectie

GROUP BY s.denumire;

```
// Număr de pompieri per secție:
SELECT s.denumire, COUNT(p.id_pompier) AS numar_pompieri
FROM Sectie s
LEFT JOIN Pompier p ON s.id_sectie = p.id_sectie
GROUP BY s.denumire;
```

Script Output | Query Result

All Rows Fetched: 5 in 0.02 seconds

	DENUMIRE	NUMAR_POMPIERI
1	Sectia Centrala	8
2	Sectia Nord	7
3	Punct de lucru Mamaia	4
4	Sectia Sud	8
5	Brasov	8

3. Pompieri care nu au participat la nicio intervenție:

SELECT nume, prenume

FROM Pompier

WHERE id_pompier NOT IN (SELECT id_pompier FROM Participare);

```
// Pompieri care nu au participat la nicio intervenție:
SELECT nume, prenume
FROM Pompier
WHERE id_pompier NOT IN (SELECT id_pompier FROM Participare);
```

Script Output x | Query Result x

SQL | All Rows Fetched: 26 in 0.023 seconds

	NUME	PRENUME
1	Manea	Claudia
2	Dobre	Ioana
3	Tudor	Cristina
4	Mihai	Bogdan
5	Manole	Paul
6	Pop	Andreea
7	Dima	Roxana
8	Ionescu	Adrian
9	Enache	Ana
10	Pruna	Dimitria
11	Petrescu	Alin
12	Georgescu	Andrei

4. Număr total de intervenții per oraș, cu număr total pompieri

SELECT s.oras,

    COUNT(DISTINCT i.id_interventie) AS nr_interventii,

    COUNT(DISTINCT pa.id_pompier) AS nr_pompieri

FROM Sectie s

JOIN Pompier p ON s.id_sectie = p.id_sectie

JOIN Participare pa ON p.id_pompier = pa.id_pompier

JOIN Interventie i ON pa.id_interventie = i.id_interventie

GROUP BY s.oras;

```sql
//Număr total de intervenții per oraș, cu număr total pompieri
SELECT s.oras,
       COUNT(DISTINCT i.id_interventie) AS nr_interventii,
       COUNT(DISTINCT pa.id_pompier) AS nr_pompieri
FROM Sectie s
JOIN Pompier p ON s.id_sectie = p.id_sectie
JOIN Participare pa ON p.id_pompier = pa.id_pompier
JOIN Interventie i ON pa.id_interventie = i.id_interventie
GROUP BY s.oras;
```

Script Output | Query Result

SQL | All Rows Fetched: 2 in 0.026 seconds

	ORAS	NR_INTERVENTII	NR_POMPIERI
1	Bucuresti	4	5
2	Constanta	4	4

5. Pompieri și numărul de intervenții la care au participat, ordonat descrescător

SELECT p.nume || ' ' || p.prenume AS pompier,

COUNT(pa.id_interventie) AS nr_interventii

FROM Pompier p

LEFT JOIN Participare pa ON p.id_pompier = pa.id_pompier

GROUP BY p.nume, p.prenume

ORDER BY nr_interventii DESC;

```sql
//Pompieri și numărul de intervenții la care au participat, ordonat descrescător
SELECT p.nume || ' ' || p.prenume AS pompier,
       COUNT(pa.id_interventie) AS nr_interventii
FROM Pompier p
LEFT JOIN Participare pa ON p.id_pompier = pa.id_pompier
GROUP BY p.nume, p.prenume
ORDER BY nr_interventii DESC;
```

Script Output | Query Result

SQL | All Rows Fetched: 35 in 0.023 seconds

	POMPIER	NR_INTERVENTII
1	Stan Elena	1
2	Neagu Mihnea	1
3	Dumitrescu Andreea	1
4	Manea Alina	1
5	Radu Alex	1
6	Ionescu Mihai	1
7	Popescu Ion	1
8	Iacob Simona	1
9	Marin Alexandru	1
10	Petrescu Alin	0
11	Enache Ana	0
12	Georgescu Andrei	0

6. Pompieri care sunt șefi și câți subordonați au

SELECT s.nume || ' ' || s.prenume AS sef,

COUNT(p.id_pompier) AS nr_subordonati

FROM Pompier s

JOIN Pompier p ON s.id_pompier = p.id_sef

GROUP BY s.nume, s.prenume

ORDER BY nr_subordonati DESC;

```
//Pompieri care sunt șefi și câți subordonați au
SELECT s.nume || ' ' || s.prenume AS sef,
       COUNT(p.id_pompier) AS nr_subordonati
FROM Pompier s
JOIN Pompier p ON s.id_pompier = p.id_sef
GROUP BY s.nume, s.prenume
ORDER BY nr_subordonati DESC;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 8 in 0.009 seconds

	SEF	NR_SUBORDONATI
1	Popescu Ion	8
2	Manea Claudia	7
3	Georgescu Andrei	6
4	Barbu Radu	3
5	Marinescu Vlad	3
6	Ionescu Mihai	3
7	Mihai Bogdan	2
8	Neagu Mihnea	2

6 queries with analytical functions

//Durata maximă și minimă a intervențiilor per secție

SELECT s.denumire AS sectie,

i.id_interventie,

i.durata_ore,

MAX(i.durata_ore) OVER (PARTITION BY s.id_sectie) AS max_durata,

MIN(i.durata_ore) OVER (PARTITION BY s.id_sectie) AS min_durata

FROM Sectie s

JOIN Pompier p ON s.id_sectie = p.id_sectie

JOIN Participare pa ON p.id_pompier = pa.id_pompier

JOIN Interventie i ON pa.id_interventie = i.id_interventie;

```
//Durata maximă și minimă a intervențiilor per secție
SELECT s.denumire AS sectie,
       i.id_interventie,
       i.durata_ore,
       MAX(i.durata_ore) OVER (PARTITION BY s.id_sectie) AS max_durata,
       MIN(i.durata_ore) OVER (PARTITION BY s.id_sectie) AS min_durata
FROM Sectie s
JOIN Pompier p ON s.id_sectie = p.id_sectie
JOIN Participare pa ON p.id_pompier = pa.id_pompier
JOIN Interventie i ON pa.id_interventie = i.id_interventie;
```

Script Output | Query Result | Query Result 1

SQL | All Rows Fetched: 9 in 0.051 seconds

	SECTIE	ID_INTERVENTIE	DURATA_ORE	MAX_DURATA	MIN_DURATA
1	Sectia Centrala	5	1.5	3.5	1.5
2	Sectia Centrala	1	3.5	3.5	1.5
3	Sectia Nord	4	2	4	2
4	Sectia Nord	1	3.5	4	2
5	Sectia Nord	6	4	4	2
6	Sectia Sud	3	7	7	3
7	Sectia Sud	7	3	7	3
8	Punct de lucru Mamaia	2	5	6	5
9	Punct de lucru Mamaia	8	6	6	5

//Rang pompieri după data angajare:

SELECT nume, prenume, data_angajare,

RANK() OVER (ORDER BY data_angajare) AS rang

FROM Pompier;

```
//Rang pompieri după data angajare:
SELECT nume, prenume, data_angajare,
       RANK() OVER (ORDER BY data_angajare) AS rang
FROM Pompier;
```

Script Output | Query Result | Query Result 1 | Query Resul

SQL | All Rows Fetched: 35 in 0.011 seconds

	NUME	PRENUME	DATA_ANGAJARE	RANG
1	Popescu	Ion	12-MAR-15	1
2	Mihai	Bogdan	12-OCT-15	2
3	Barbu	Radu	16-MAY-16	3
4	Neagu	Mihnea	20-JUN-16	4
5	Manea	Claudia	25-MAY-17	5
6	Marinescu	Vlad	19-SEP-17	6
7	Popa	Cristina	17-MAR-18	7
8	Ionescu	Mihai	10-MAY-18	8
9	Vasilescu	Andrei	27-FEB-19	9
10	Constantinescu	Elena	22-APR-19	10
11	Georgescu	Andrei	15-SEP-19	11
12	Stan	Elena	01-NOV-19	12

//Durata totală a intervențiilor cumulative:

SELECT id_interventie, durata_ore,

SUM(durata_ore) OVER (ORDER BY id_interventie) AS durata_cumulativa

FROM Interventie;

```
//Durata totală a intervențiilor cumulative:
SELECT id_interventie, durata_ore,
       SUM(durata_ore) OVER (ORDER BY id_interventie) AS durata_cumulativa
FROM Interventie;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 15 in 0.01 seconds

	ID_INTERVENTIE	DURATA_ORE	DURATA_CUMULATIVA
1	1	3.5	3.5
2	2	5	8.5
3	3	7	15.5
4	4	2	17.5
5	5	1.5	19
6	6	4	23
7	7	3	26
8	8	6	32
9	9	5	37
10	10	2.5	39.5
11	11	3.5	43
12	12	1	44
13	13	4	48

//Rang pompieri pe secție după grad:

SELECT nume, prenume, grad, id_sectie,

RANK() OVER (PARTITION BY id_sectie ORDER BY grad DESC) AS rang_grad

FROM Pompier;

```
//Rang pompieri pe secție după grad:
SELECT nume, prenume, grad, id_sectie,
       RANK() OVER (PARTITION BY id_sectie ORDER BY grad DESC) AS rang_grad
FROM Pompier;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Quer

SQL | All Rows Fetched: 35 in 0.011 seconds

	NUME	PRENUME	GRAD	ID_SECTIE	RANG_GRAD
1	Popa	Cristina	Sergent	1	1
2	Vasilescu	Andrei	Sergent	1	1
3	Manole	Paul	Pompier	1	3
4	Petrescu	Alin	Pompier	1	3
5	Enache	Ana	Pompier	1	3
6	Neagu	Mihnea	Locotenent	1	6
7	Mihai	Bogdan	Capitan	1	7
8	Popescu	Ion	Capitan	1	7
9	Stan	Elena	Sergent	2	1
10	Alexandru	Laura	Pompier	2	2
11	Dumitrescu	Andreea	Pompier	2	2
12	Dobre	Ioana	Pompier	2	2
13	Cojocaru	Robert	Pompier	2	2

//Diferență față de participarea anterioară (LAG)

SELECT id_interventie,

durata_ore,

LAG(durata_ore, 1) OVER (ORDER BY data_interventie) AS durata_precedenta,

durata_ore - LAG(durata_ore, 1) OVER (ORDER BY data_interventie) AS diferenta

FROM Interventie;

```
//Diferență față de participarea anterioară (LAG)
SELECT id_interventie,
       durata_ore,
       LAG(durata_ore, 1) OVER (ORDER BY data_interventie) AS durata_precedenta,
       durata_ore - LAG(durata_ore, 1) OVER (ORDER BY data_interventie) AS diferenta
FROM Interventie;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 ×

SQL | All Rows Fetched: 15 in 0.009 seconds

	ID_INTERVENTIE	DURATA_ORE	DURATA_PRECEDENTA	DIFERENTA
1	1	3.5	(null)	(null)
2	2	5	3.5	1.5
3	3	7	5	2
4	4	2	7	-5
5	5	1.5	2	-0.5
6	6	4	1.5	2.5
7	7	3	4	-1
8	8	6	3	3
9	9	5	6	-1
10	10	2.5	5	-2.5
11	11	3.5	2.5	1
12	12	1	3.5	-2.5
13	13	4	1	3

```
//Rangul intervențiilor după durată
SELECT id_interventie,
    durata_ore,
    RANK() OVER (ORDER BY durata_ore DESC) AS rang_durata
FROM Interventie;
```

```sql
//Rangul intervențiilor după durată
SELECT id_interventie,
       durata_ore,
       RANK() OVER (ORDER BY durata_ore DESC) AS rang_durata
FROM Interventie;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 15 in 0.01 seconds

	ID_INTERVENTIE	DURATA_ORE	RANG_DURATA
1	3	7	1
2	8	6	2
3	15	5.5	3
4	2	5	4
5	9	5	4
6	6	4	6
7	13	4	6
8	1	3.5	8
9	11	3.5	8
10	7	3	10
11	10	2.5	11
12	14	2	12
13	4	2	12